

Table S4: Operationalization of the *Landsat-to-Global Freshwater Intelligence Framework*

Pillar	Critical research gap	Critical dependencies	Testable proposition	Dependency-based maturity milestone	Proposed measurable indicators of progress
P1. Interoperable Earth Observation Infrastructure	Full operational interoperability across missions remains limited by differences in preprocessing, atmospheric correction, radiometric harmonization, and uncertainty characterization, particularly for optically complex inland waters.	Foundational/enabling pillar. Provides the harmonized observations and scalable processing infrastructure required by P2–P5.	P1. Standardized preprocessing and radiometrically harmonized multi-sensor observations should increase the consistency and reproducibility of water-quality time series relative to mission-specific processing chains.	M1 — Interoperability: harmonized and reproducible observations can be generated consistently across sensors, regions, and time, providing the observational basis for M2–M5.	Cross-sensor bias and uncertainty; reproducibility across processing platforms; temporal observation density; spatial coverage of interoperable products; adoption of standardized/ARD-compatible processing.
P2. From Local Models to Global Water Intelligence	Benchmark datasets remain geographically, limnologically, and optically uneven, limiting independent evaluation and the transition beyond the “ one lake, one model ” paradigm.	Requires M1 for standardized observations. Globally representative benchmark/reference datasets are a critical prerequisite for training and independently validating P3 models.	P2. Increasing the geographic, limnological, and optical diversity of benchmark datasets should reduce performance degradation when retrieval algorithms are transferred to previously unseen water bodies and regions.	M2 — Generalization: sufficiently representative and harmonized reference datasets enable multi-regional benchmarking, independent validation, and transferable retrieval approaches. M1 + M2 enable M3.	Number and geographic distribution of satellite–in situ matchups; representation of OWTs and climatic/limnological regions; performance degradation between calibration and independent regions; number of algorithms evaluated using common benchmarks; demonstrated out-of-region transferability.
P3. Physics-Informed Artificial Intelligence (PIAI)	Data-driven ML may lose performance outside its training domain and provide limited physical interpretability; spectral ambiguities and insufficient uncertainty characterization constrain robust transferability.	Requires M1 + M2 for interoperable inputs and representative training/validation data. Interacts bidirectionally with P4, as enhanced spectral information strengthens physical constraints while PIAI provides methods for exploiting these observations.	P3. Incorporating radiative-transfer, bio-optical, and IOP constraints into AI models should improve physical consistency and out-of-domain transferability relative to equivalent unconstrained data-driven models.	M3 — Transferability: retrieval models demonstrate out-of-domain robustness, physical plausibility, characterized uncertainty, and reproducibility in addition to predictive performance.	In- versus out-of-domain RMSE/bias/R ² ; performance degradation outside the training domain; frequency of physically implausible retrievals; uncertainty calibration; performance across OWTs/regions; reproducibility.
P4. Landsat Next and Bio-optical Characterization	Current multispectral capabilities constrain discrimination of spectrally overlapping constituents and retrieval of IOPs, particularly in optically complex waters.	Builds on M1 and M3 . P3 ↔ P4 represents a bidirectional relationship: enhanced spectral observations strengthen PIAI/bio-optical modelling, while physics-informed approaches enable more effective exploitation of next-generation observations.	P4. Integrating Landsat temporal continuity with spectrally richer multispectral/hyperspectral observations should improve constituent discrimination and IOP retrieval while preserving continuity with the historical Landsat record.	M4 — Characterization: interoperable bio-optical products based increasingly on IOPs and physically consistent constituent characterization are linked to the multi-decadal Landsat record.	Accuracy and uncertainty of IOP retrievals; discrimination of aquatic constituents; cross-sensor consistency; improvement relative to current Landsat retrievals; performance in optically complex waters; continuity of derived products.
P5. From Scientific Products to Environmental Governance	Satellite products remain insufficiently standardized, uncertainty-characterized, and integrated with in situ observations, environmental models, and institutional reporting for sustained routine use.	Downstream/integration pillar. Requires sufficiently mature outcomes from M1–M4. Operational implementation also generates validation, uncertainty, and user requirements that feed back to P1–P4 .	P5. Standardized, transparent, and uncertainty-characterized EO products integrating satellite and in situ information should complement conventional monitoring and increase the spatial and temporal coverage of environmental assessment and reporting.	M5 — Operational Intelligence: sustained integration of EO, in situ observations, models, and institutional reporting produces decision-grade freshwater intelligence for environmental governance.	Number of operational/regulatory implementations; traceability of uncertainty across the processing chain; agreement with regulatory in situ assessments; reporting frequency and geographic coverage; uptake in SDG 6.3.2/WFD or comparable frameworks; continuity of operational services.

The table operationalizes the five pillars of the proposed framework by linking each pillar to the conditions required for its progressive maturation. For each pillar, the **Critical research gap** identifies the main limitation that remains to be addressed, while **Critical dependencies** describe how progress depends on, or contributes to, the development of the other pillars. The **Testable proposition** translates the expected advance into a hypothesis that can be empirically evaluated, and the **Dependency-based maturity milestone** defines the condition that would indicate that the pillar has reached a sufficient level of development to enable subsequent stages of the framework. Finally, the **Proposed measurable indicators** of progress provide practical criteria for assessing whether such progress is occurring. The milestones are therefore not intended as fixed chronological stages, but as interdependent levels of maturity, with feedback among pillars allowing new operational requirements and scientific advances to continuously refine the framework.